

2. Three connected tanks

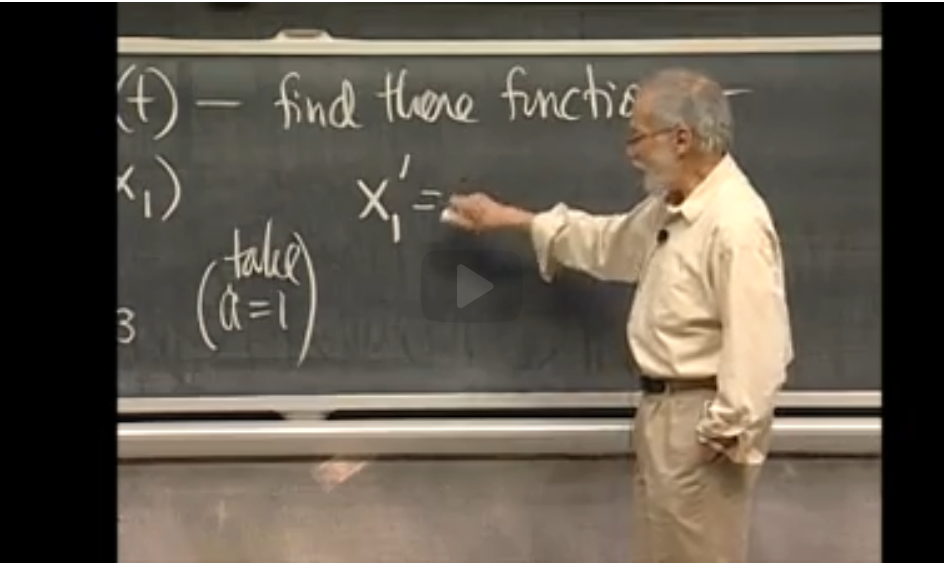
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Three cyclically connected fish tanks



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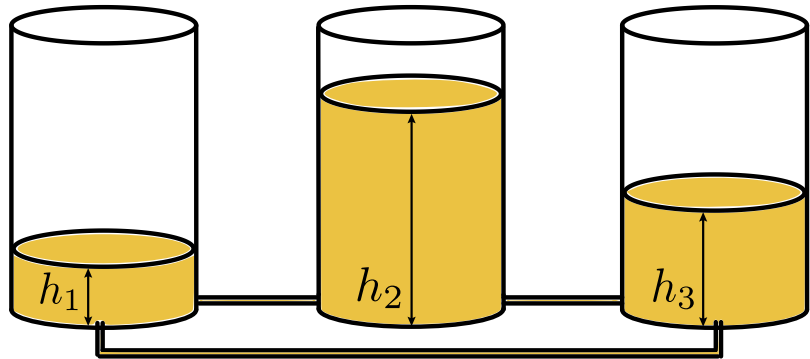
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x1 prime is-- and let's take a equal 1, because I'm going to want to solve them numerically and I want you to be able to concentrate on what's important. You know, the what's New, now. I'm not too fussed with the-- I don't want to have an extra a floating around everywhere just contributing nothing but a mild confusion to the proceedings. So X1 prime-- I'm going to take a equal one, and simply write it minus 2x1 plus x2 plus x3. And so now what would the equation for x2 prime be? Well here, x2 plays the role that x1 played before. And the only way to tell you that x1 was

Below we will go through another example which yields exactly the same system of differential equations as in the video.

You've been put in charge of part of a medicine manufacturing plant with three identical cylindrical storage tanks. These tanks are filled with the medicine to different heights and are connected to each other via a pipe at their bottoms, with valves that are initially closed. Our goal is to determine how the fluid will flow once we open the valves. Note that we have solved the analogous problem for two tanks in the course *Differential equations: 2 by 2 systems* previously.

To begin, we name the heights of the fluid in the three tanks h_1 , h_2 , and h_3 respectively.



Three tanks cyclically connected to each other

Simplifying assumptions

We will make the simplifying assumptions that we can model the flow linearly, and that the flow rate is proportional to the difference in fluid height (as in the case with only 2 tanks). These assumptions are valid when the pipes have a small diameter relative to the overall size of the tanks, the geometry of the pipes and tanks is simple, and the fluid properties of the medicine work out.

Equation for tank 1:

Let us first work out the differential equation describing fluid flow into tank 1. Let V_1 be the volume of fluid in tank 1. The flow rate of fluid into (or out of) tank 1 is by definition $\frac{dV_1}{dt}$.

As mentioned above, we assume that the flow rate is proportional to the difference in heights. The flow rate into tank 1 from tank 2 is $b(h_2 - h_1)$, where b is the constant of proportionality (with dimension $[\text{length}]^2[\text{time}]^{-1}$). Similarly, the flow rate from tank 3 is $c(h_3 - h_1)$, with c the constant of proportionality. The flow rate into tank 1 is therefore:

$$\frac{dV_1}{dt} = b(h_2 - h_1) + c(h_3 - h_1).$$

(4.1)

Note that the constant b is positive because when $h_2 > h_1$, fluid flows into tank 1 so that V_1 is increasing. Similarly, c is positive. We usually set up differential equations so that parameters such as b and c are positive. This way we will be less prone to sign errors.

Since the tank is cylindrical, the rate of change in volume V_1 is given by the cross sectional area A times the rate of change in height h_1 of fluid in tank 1 :

$$\frac{dV_1}{dt} = A \frac{dh_1}{dt}.$$

(4.2)

Therefore, the differential equation in terms of h_1 is

$$\frac{dh_1}{dt} = \frac{1}{A} \frac{dV_1}{dt} = \frac{b}{A}(h_2 - h_1) + \frac{c}{A}(h_3 - h_1). \tag{4.3}$$

We can combine the constants. Let $a_{21} = \frac{b}{A}$ (with dimension $[\text{time}]^{-1}$), where the subscript **21** indicates that this is the constant governing the flow from tank 2 to tank 1. Similarly, let $a_{31} = \frac{c}{A}$ be the combined constant governing flow from tank 3 and tank 1. The differential equation in terms of the new constants is:

$$\frac{dh_1}{dt} = a_{21} (h_2 - h_1) + a_{31} (h_3 - h_1) \quad (a_{21}, a_{31} > 0). \tag{4.4}$$

Equations for tank 2 and tank 3:

Analogous reasoning as above gives the differential equations for flow rates into tank 2 and tank 3:

$$\frac{dh_2}{dt} = a_{12} (h_1 - h_2) + a_{32} (h_3 - h_2) \quad (a_{12}, a_{32} > 0) \tag{4.5}$$

$$\frac{dh_3}{dt} = a_{13} (h_1 - h_3) + a_{23} (h_2 - h_3) \quad (a_{13}, a_{23} > 0). \tag{4.6}$$

Note that $a_{12} = a_{21}$. For example, if $h_2 > h_1$, then $a_{21} (h_2 - h_1) > 0$ so the flow rate from tank 2 into tank 1 is positive, whereas $a_{12} (h_1 - h_2) < 0$ so that the flow rate from tank 1 to tank 2 is negative. This is consistent because these 2 expressions are equal in magnitude but opposite in sign. Similarly, $a_{13} = a_{31}$, and $a_{23} = a_{32}$.

Matrix form

We can now rewrite this equation in matrix form:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} \quad \text{where } \mathbf{x} = \begin{pmatrix} h_1 \\ h_2 \\ h_3 \end{pmatrix}, \mathbf{A} = \begin{pmatrix} -a_{12} - a_{13} & a_{12} & a_{13} \\ a_{12} & -a_{12} - a_{23} & a_{23} \\ a_{13} & a_{23} & -a_{13} - a_{23} \end{pmatrix}. \tag{4.7}$$

To make things even nicer, we'll set all constants $a_{ij} = 1$. The resulting equation is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} \quad \text{where } \mathbf{x} = \begin{pmatrix} h_1 \\ h_2 \\ h_3 \end{pmatrix}, \mathbf{A} = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}. \tag{4.8}$$

Symmetry

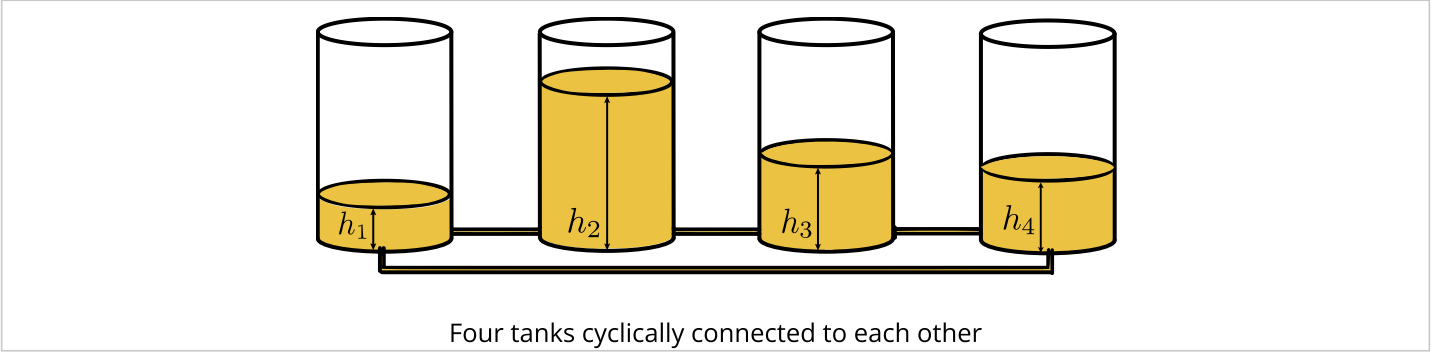
Note that the matrix $\mathbf{A}$ is symmetric, i.e. $\mathbf{A} = \mathbf{A}^T$ (even before the constants are set to **1**). Is this a coincidence? In fact, symmetric matrices appear in many models of real physical phenomena.

An important theorem in linear algebra states that symmetric matrices have real eigenvalues and an orthogonal basis of eigenvectors. We will see how these properties are useful in the next lecture.

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Four cyclically connected tanks

1/1 point (graded)



Consider the setup as in the example above but with 4 cyclically-connected tanks instead. As in the example above, assume the flow rate constants (with dimension $[\text{time}]^{-1}$) along each of the 4 pipes are set to **1**, and let h_i be the height of fluid in tank i .

Find the matrix $\mathbf{A}$ such that the system of DEs describing the fluid flow between the tanks is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} \quad \text{where } \mathbf{x} = \begin{pmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \end{pmatrix}. \tag{4.9}$$

(Enter `[[a,b],[c,d]]` for the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. That is, use commas to separate entries within the same row and then enclose the entire row in **square brackets**. Then use **commas** to separate rows, and **square bracket** the entire matrix. Similarly for larger matrices.)

A =

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[[-2,1,0,1],[1,-2,1,0],[0,1,-2,1],[1,0,1,-2]]



Answer: [[-2,1,0,1],[1,-2,1,0],[0,1,-2,1],[1,0,1,-2]]

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$$\frac{dh_2}{dt} = (h_1 - h_2) + (h_3 - h_2) \tag{4.11}$$

$$\frac{dh_3}{dt} = (h_2 - h_3) + (h_4 - h_3) \tag{4.12}$$

$$\frac{dh_4}{dt} = (h_3 - h_4) + (h_1 - h_4) . \tag{4.13}$$

Hence, the matrix **A** is

$$\mathbf{A} = \begin{pmatrix} -2 & 1 & 0 & 1 \\ 1 & -2 & 1 & 0 \\ 0 & 1 & -2 & 1 \\ 1 & 0 & 1 & -2 \end{pmatrix} . \tag{4.14}$$

Put another way, in the **3 × 3** case:

$$\frac{dh_1}{dt} = \begin{pmatrix} -2 & 1 & 1 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \\ h_3 \end{pmatrix} . \tag{4.15}$$

In the **4 × 4** case, since there is no connection between tanks 1 and 3, **a₃₁ = 0**, and we get

$$\frac{dh_1}{dt} = \begin{pmatrix} -2 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \end{pmatrix} . \tag{4.16}$$

Similarly, the other equations are determined by the **3 × 3** case and the fact that **a₁₃ = a₃₁ = a₂₄ = a₄₂ = 0**.

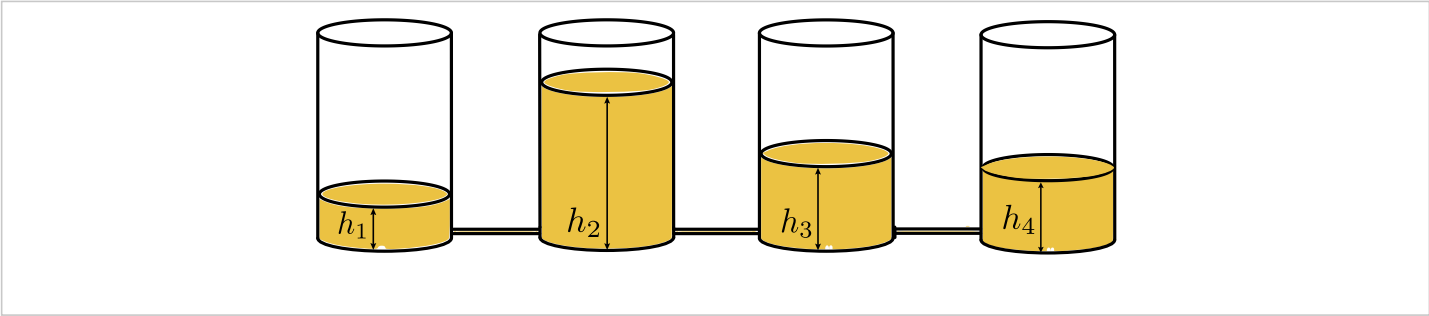
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Four linearly connected tanks

1/1 point (graded)



As above, consider four connected fluid tanks. But this time, tank 1 and tank 4 are **not connected**.

As in the example above, assume the flow rate constant (with dimension **[time]⁻¹**) along each of the 4 pipes are set to **1**, and let **h_i** be the height of fluid in tank **i**.

Find the matrix **A** such that the system of DE describing the fluid flow between the tanks is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} \quad \text{where } \mathbf{x} = \begin{pmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \end{pmatrix} . \tag{4.17}$$

(Enter **[[a,b],[c,d]]** for the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. That is, use commas to separate entries within the same row and then enclose the entire row in **square brackets**. Then use **commas** to separate rows, and **square bracket** the entire matrix. Similarly for larger matrices.)

A =

[[-1,1,0,0],[1,-2,1,0],[0,1,-2,1],[0,0,1,-1]]



Answer: [[-1,1,0,0],[1,-2,1,0],[0,1,-2,1],[0,0,1,-1]]

Solution:

The only differential equations that are changed are the ones for $\frac{dh_1}{dt}$ and $\frac{dh_4}{dt}$.
In this setup, fluid flows into tank 1 only from tank 2, and hence

$$\frac{dh_1}{dt} = (h_2 - h_1) . \tag{4.18}$$

Similarly, fluid now flows into tank 4 only from tank 3, so

$$\frac{dh_4}{dt} = (h_3 - h_4) . \tag{4.19}$$

The differential equations for $\frac{dh_2}{dt}$ and $\frac{dh_3}{dt}$ remain the same as in the previous problem. Together these DEs form the system

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} \qquad \text{where } \mathbf{x} = \begin{pmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \end{pmatrix} \text{ and } \mathbf{A} = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} . \tag{4.20}$$

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2. Three connected tanks

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